

AWS EC2 – Application Load Balancer with Two EC2 Instances (Full Practical)

Task:

Create two EC2 instances, install Nginx on both, and configure an Application Load Balancer to distribute traffic between them.

(a) Create Two EC2 Instances:

1. Login to AWS Management Console
Open <https://aws.amazon.com/> and sign in
2. Open EC2 Dashboard
Go to Services → EC2
3. Click Launch Instance
4. Create First Instance:
 - Name: WebServer1
 - AMI: Amazon Linux 2 / Ubuntu 22.04
 - Instance Type: t2.micro
 - Key Pair: mykey.pem
 - Security Group:
 - Allow SSH (22)
 - Allow HTTP (80)
 - Click Launch
5. Create Second Instance:
 - Name: WebServer2
 - Same configuration as above
 - Click Launch
6. Wait until both instances are Running

(b) Install Nginx on Both Instances:

Connect:

```
ssh -i mykey.pem ec2-user@<Public-IP>
Install Nginx (Amazon Linux):
sudo yum update -y
sudo amazon-linux-extras install nginx1 -y
sudo systemctl start nginx
sudo systemctl enable nginx
```

OR (Ubuntu):

```
sudo apt update -y
sudo apt install nginx -y
sudo systemctl start nginx
sudo systemctl enable nginx
```

(c) Create Custom Pages:

On WebServer1:

```
echo "<h1>Server 1</h1>" | sudo tee /usr/share/nginx/html/index.html
```

On WebServer2:

```
echo "<h1>Server 2</h1>" | sudo tee /usr/share/nginx/html/index.html
```

(d) Create Target Group:

Go to EC2 Dashboard

Click Target Groups

Click Create Target Group

Set:

Target Type: Instances

Name: MyTargetGroup

Protocol: HTTP

Port: 80

Click Next

Select both instances

Click Create Target Group

(e) Create Application Load Balancer:

Go to EC2 Dashboard

Click Load Balancers

Click Create Load Balancer

Choose Application Load Balancer

Configure:

Name: MyALB

Scheme: Internet-facing

IP type: IPv4

Network:

Select VPC

Select at least 2 subnets

Security Group:

Allow HTTP (Port 80)

Listener:

HTTP (Port 80)

Select Target Group:

MyTargetGroup

Click Create Load Balancer

(f) Test Load Balancer:

Copy DNS name of ALB

Example:

http://myalb-123456.ap-south-1.elb.amazonaws.com

Open in browser

Refresh multiple times:

Output alternates between:

Server 1

Server 2

Result:

Two EC2 instances created, Nginx installed, ALB configured, and traffic successfully distributed